From Long Term Classroom Programs and Medium / Short Classroom Program 4 in Top 10, 10 in Top 20, 43 in Top 100, 75 in Top 200, 159 in Top 500 Ranks & 3542 total selections in IIT-JEE 2012

ALL INDIA TEST SERIES

FIITJEE

JEE (Advanced), 2013

FULL TEST – IV

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 210**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into two sections: **Section-A & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 10)** contains 10 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (11 to 15) contains 5 multiple choice questions which have more than one correct answer. Each question carries **+4 marks** for correct answer. There is no negative marking.

(ii) **Section-C (01 to 05)** contains 5 Numerical based questions with answers as numerical value and each question carries **+4 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

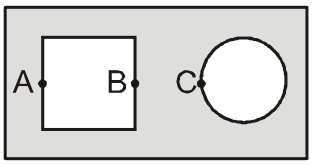
PART – I

SECTION – A

Single Correct Choice Type

This section contains **10 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

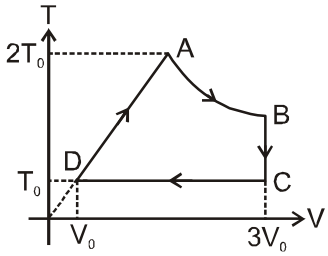
1. Two large holes are cut in a metal sheet as shown. If this is heated, which distance will decrease :



(A) BC (B) AB
(C) AC (D) None of these

2. 2 gms steam at 100° is mixed with 5 gm ice at – 40°C in an ideal calorimeter. The final temperature of the system will be (Given L_v = 500 cal/gm; S_{ice} = 0.5 cal/gm°C, S_w = 1 cal/gm °C; L_f = 80 cal/gm) :
(A) 0°C (B) 100°C
(C) 80°C (D) 50°C

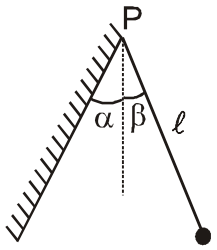
3. Consider the cyclic process ABCDA, as shown in figure, performed on a sample of n moles of an ideal gas. Net heat supplied to the gas during the process is $4nRT_0$. Then the work done during the process AB will be :



(A) $3nRT_0$
(B) $nRT_0 (4 + \ell n3)$
(C) $nRT_0 (3 + \ell n3)$
(D) $nRT_0 (2 + \ell n3)$

Rough work

4. A ball is hung vertically by a thread of length '' from a point 'P' of an inclined wall that makes an angle 'α' with the vertical. The thread with the ball is then deviated through a small angle 'β' ($\alpha > \beta$) and set free. Assuming the wall to be perfectly elastic, the period of such pendulum is :



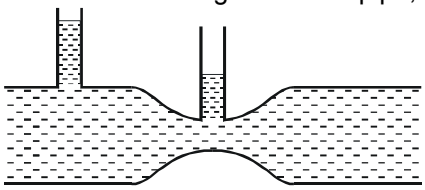
(A) $2\sqrt{\frac{\ell}{g}}\left[\sin^{-1}\left(-\frac{\alpha}{\beta}\right)\right]$

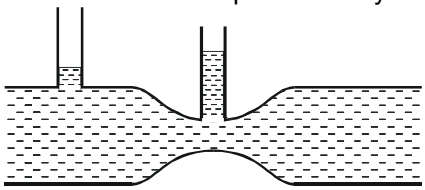
(B) $2\sqrt{\frac{\ell}{g}}\left[\frac{\pi}{2}+\sin^{-1}\left(\frac{\alpha}{\beta}\right)\right]$

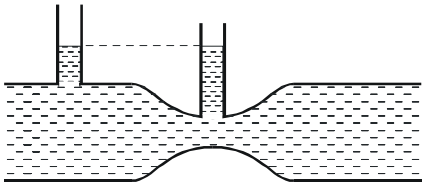
(C) $\pi\sqrt{\frac{\ell}{g}}$

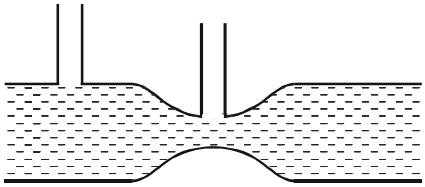
(D) $2\pi\sqrt{\frac{\ell}{g}}$

5. For a static fluid in a long horizontal pipe, the level in the vertical tubes is best represented by :

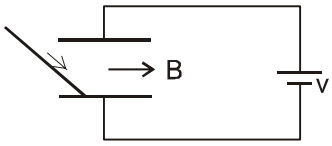
(A) 

(B) 

(C) 

(D) 

6. The collector plate in an experiment on photoelectric effect is kept vertically above the emitter plate. Light source is put on and a saturation photocurrent is recorded. A magnetic field is switched on which is in horizontal rightward direction :



(A) The photocurrent will increase (B) The photocurrent may decrease
(C) The photocurrent will always remain same (D) None of these

Rough work

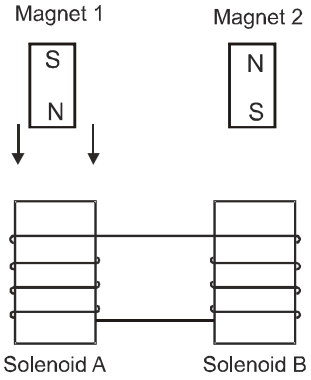
7. An uncharged capacitor of capacitance C is connected with an ideal cell. The emf of the cell is slowly increased from 0 to V (by some mechanism) . The total energy taken from the cell in the process of charging of the capacitor is (assume the resistance of the circuit is very small):

(A) $\frac{1}{2}CV^2$ (B) $2CV^2$

(C) $\frac{1}{4}CV^2$ (D) CV^2

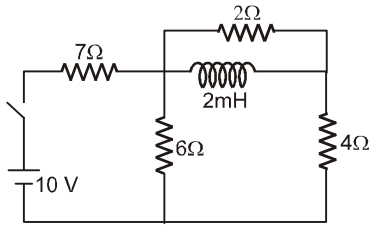
8. Two hollow-core solenoids, A and B, are connected by a wire and separated by a large distance, as shown in the diagram. Two bar magnets, 1 and 2, are suspended just above the solenoids. If the magnet 1 is dropped through solenoid A as shown, then the magnet 2 will simultaneously be

(A) Attracted by a magnetic force towards solenoid B
(B) Repelled by a magnetic force away from solenoid B
(C) Repelled by an electric force away from solenoid B
(D) Unaffected by solenoid B.



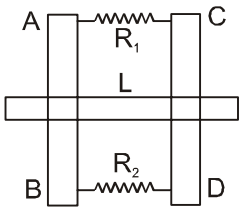
9. In the given circuit find the ratio of i_1 to i_2. Where i_1 is the initial (at t = 0) current, and i_2 is steady state (at t = ∞) current through the battery :

(A) 1.0 (B) 0.8
(C) 0.94 (D) 1.06



10. Two parallel vertical metallic rails AB and CD are separated by 1 m. They are connected at the two ends by resistance R_1 and R_2 as shown in the figure. A horizontal metallic bar L of mass 0.2 kg slides without friction, vertically down the rails under the action of gravity. There is a uniform horizontal magnetic field of 0.6T perpendicular to the plane of the rails. It is observed that when the terminal velocity of 4 m/s is attained, power dissipated in R_1 is 2 W. Then the power dissipated in R_2 at this moment will be :

(A) 2 W (B) 4 W
(C) 8 W (D) 6 W



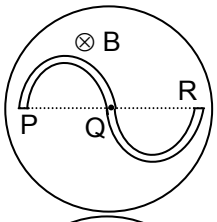
Rough work

Multiple Correct Answer(s) Type

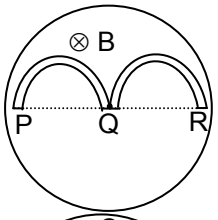
This section contains **5 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE are correct**.

11. In each of the following diagrams, the magnetic field in the circular region is inside the plane of the paper and is increasing with time. Four different shaped metallic conductors 'PQR' are placed as shown. The end P will be more positively charged compared to R in: (Q is the centre in options AB).

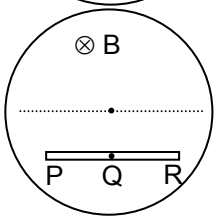
(A)



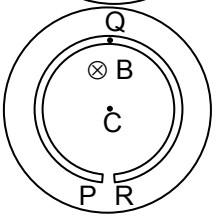
(B)



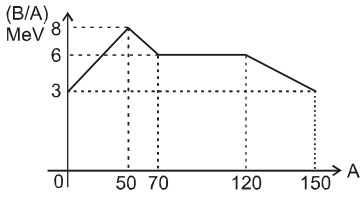
(C)



(D)



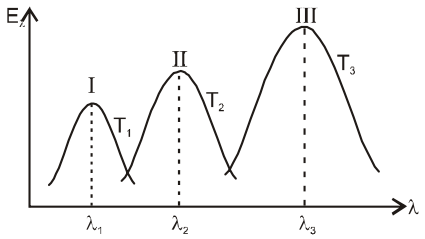
12. Assume that the nuclear binding energy per nucleon (B/A) versus mass number (A) is as shown in the figure. Consider a nucleus of A = 110. Fission of this nucleus results into 2 fragments. Which of the following could possibly be the mass number of the resulting nuclei?



(A) 55 and 55
(B) 70 and 40
(C) 100 and 10
(D) 90 and 20

Rough work

13. Three identical bodies are at temperature T_1, T_2 and T_3 having e_1, e_2 and e_3 as their respective emissivities. The thermal spectrum obtained for them is as shown in the diagram. Choose the correct order of temperatures and emissivities:



(A) $T_1 > T_2 > T_3$
(B) $T_1 < T_2 < T_3$
(C) $e_1 < e_2 < e_3$
(D) $e_1 > e_2 > e_3$

14. When photons of energy 4.25 eV strike the surface of a metal A, the ejected photoelectrons have maximum kinetic energy, T_A expressed in eV and de Broglie wavelength λ_A. The maximum kinetic energy of photoelectrons liberated from another metal B by photons of energy 4.70 eV is $T_B = (T_A - 1.50 \text{ eV})$. If the de Broglie wavelength of these photoelectrons is $\lambda_B = 2\lambda_A$, then :
(A) the work function of A is 2.25 eV
(B) the work function of B is 4.20 eV
(C) $T_A = 2.00$ eV
(D) $T_B = 2.75$ eV

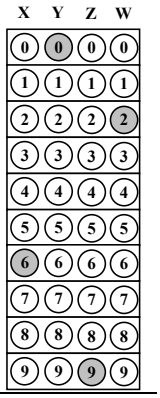
15. An ideal monoatomic gas undergoes a process in which $P = 3V^3$. Choose the correct statements.
(A) If volume increases then temperature will also increase
(B) If temperature of the gas decreases, then pressure must also decrease.
(C) For expansion, heat will have to be supplied to the gas.
(D) If temperature increases then work done by gas is positive.

Rough work

SECTION – C

Integer Answer Type

This section contains 5 questions. The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown.



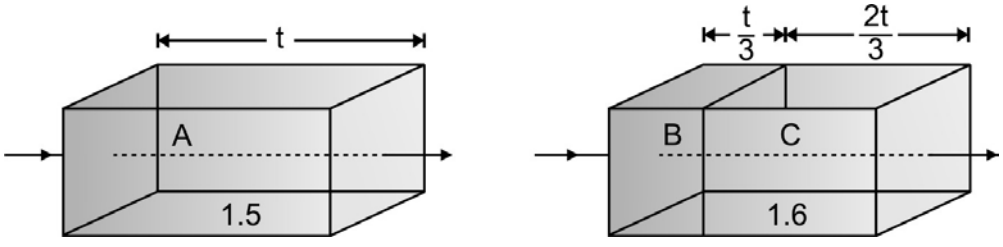
1. A string of length 2L, obeying Hooke's Law, is stretched so that its extension is L. The speed of the tranverse wave travelling on the string is v. If the string is further stretched so that the extension in the string becomes 4L. The speed of transverse wave travelling on the string will be $x\sqrt{2}\,v$. Calculate the value of x.

2. A train moves towards a stationary observer with a speed of 34 ms^{-1}. The train sounds a whistle and its frequency registered by the observer is f_1. If the train's speed is reduced to 17 ms^{-1}, the frequency registered is f_2. If the speed of sound is 340 ms^{-1}, then the ratio $\frac{f_1}{f_2}$ is $\frac{10+x}{18}$. Calculate the value of x.

3. A tuning fork A of frequency as given by the manufacturer is 512 Hz is being tested using an accurate oscillator. It is found that they produce 2 beats per second when the oscillator reads 514 Hz and 6 beats per second when it reads 510 Hz. The actual frequency of the fork in Hz is (510 + x). Calculate the value of x.

4. A person runs with a speed u towards a bicycle moving away from him with speed v. The person approaches his image in the mirror fixed at the rear bicycle with a speed of x(u – v). Calculate the value of x.

5. Two transparent slabs have the same thickness as shown. One is made of material A of refractive index 1.5. The other is made of two materials B and C with thickness in the ratio 1 : 2. The refractive index of C is 1.6. If a monochromatic parallel beam passing through the slabs has the same number of wavelengths inside both, the refractive index of B is $\frac{10+x}{10}$. Then x is:



Rough work

Chemistry

PART – II

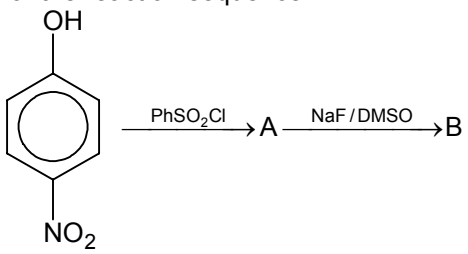
SECTION – A

Straight Objective Type

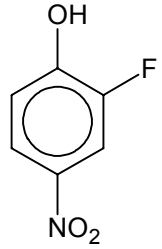
This section contains 10 multiple choice questions numbered 1 to 10. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

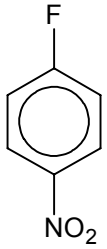
1. A compound (X) has the molecular formula C_3H_7NO gives Y on reaction with KOH and Br_2. Compound Y responds to mustard oil reaction. Y upon treatment with HNO_2 evolves N_2 and gives an alcohol Z which gives iodoform test. The compound X can be
   (A) $C_2H_5CONH_2$ (B) CH_3CONH_2
   (C) CH_3COONH_4 (D) C_2H_5NO

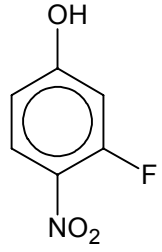
2. For the reaction sequence

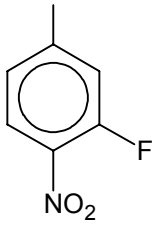


The compound B will be

(A) 

(B) 

(C) 

(D) 

3. In an experiment 11 g PCl_3 is made to react with 1.34 g of O_2 to yield $POCl_3$. If 11.2 g of $POCl_3$ were isolated. What is the percentage yield.
   (A) 60 (B) 70
   (C) 80 (D) 90

4. What is the molar mass of the gas if the density of a gas sample is 0.714 g/L at STP?
   (A) 24 (B) 16
   (C) 36 (D) 42

Rough Work

5. What is the root mean square speed of Argon atoms at 27°C.
(A) 200 m/s (B) 341 m/s
(C) 433 m/s (D) 510 m/s

6. Find the wavelength of light emitted by an electron during a transition from n = 3 to n = 1 level in the C^{+5} ion. Given Rh = 1.097×10^7 m^{-1}.
(A) 2.85 nm (B) 8.76 nm
(C) 6.40 nm (D) 12.0 nm

7. What is the shape of ClNO
(A) V- shaped (B) T – shaped
(C) Triangle (D) Linear

8. NCl_3 is hydrolysed by water to produce
(A) NH_2NH_2 and HCl (B) NH_2Cl + HOCl
(C) NH_4OH + HOCl (D) NH_2OH + HCl

9. Select the incorrect choice :
(A) solubility of alkaline earth metal's carbonates, sulphates and chromates decreases from Be to Ba.
(B) solubility of alkaline earth metal's hydroxides is less than alkali metal hydroxides.
(C) solubility of alkaline earth metal's oxides increases from Be to Ba.
(D) SO_2 on passing in lime water turns is milky.

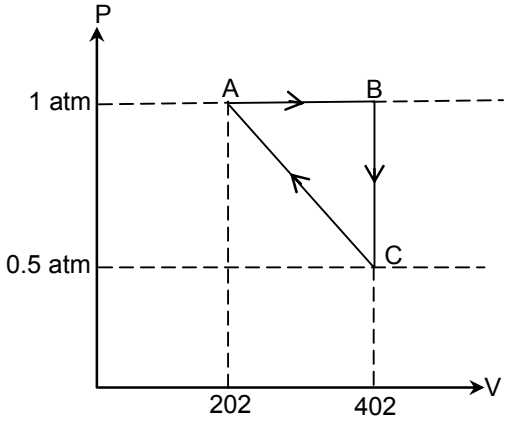
10. The reduction potential of Cu^+ | Cu half cell if SRP of Cu^+ | Cu and SRP of Cu^{2+}| Cu^+ are 0.34 and 0.16 V respectively, will be
(A) 0.36 V (B) 0.43 V
(C) 0.68 V (D) 0.52 V

Rough Work

Multiple Correct Choice Type

This section contains 5 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct.

11. On the basis of the following graph (P-V) graph, choose the correct statements.



(A) The entropy change for the over all process is zero
(B) For the over all process $\Delta H > \Delta E$
(C) Total work done, W = q
(D) Total work = – 620.77 J

12. 150 ml $\frac{M}{10}$ Ba $(MnO_4)_2$ in acidic medium can oxidise completely.

(A) 150 ml of 1 M Fe^{+2}
(B) 50 ml of 1 M $FeCrO_4$
(C) 25 ml of 1 M $K_2Cr_2O_7$
(D) 100 ml of 2 M $C_2O_4^{-2}$ ions

13. Which of the following is/are produced during heating of Gypsum?

(A) CaO
(B) $Ca(OH)_2$
(C) $CaSO_4$
(D) Plaster of Paris

14. Which of the following solution will be acidic

(A) 0.1 M CH_3COONa
(B) 0.1 M $FeSO_4$
(C) 0.1 M $K_2SO_4Al_2(SO_4)_3.24H_2O$
(D) 0.1 M $(NH_4)_2SO_4$

15. If temperature of a gas is raised, which of the following would be true?

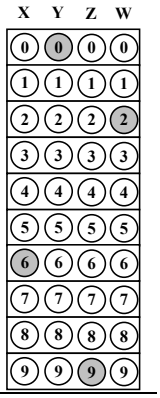
(A) Fraction of the molecules possessing most probable velocity will increase
(B) Fraction of the molecules possessing most probable velocity will decrease
(C) Fraction of possessing very low velocity will decrease
(D) Fraction possessing very high velocity will increase

Rough Work

SECTION – C

Integer Answer Type

This section contains 5 questions. The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown.



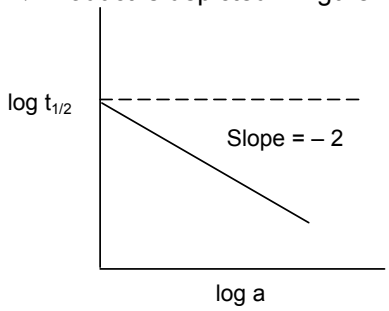
1. An organic compound (A) (m.f. $C_{10}H_{20}$) on reductive ozonolysis gives 2-methyl butanal. Then total no. of steroisomers possible for A is

2. If the half life of a radioactive sample is 2n years. Then the fraction of a sample which will remain decayed after n years is $\frac{1}{\sqrt{x}}$. Then the value of x will be

3. A mixture containing x millimoles of NaOH and y milli moles of Na_2CO_3 requires.
   (A) 4 ml of 1 N HCl using phenolphthalien indicator
   (B) 5 ml of 1 N HCl using methyl orange indicator
   The value of x/y will be

Rough Work

4. A graph between log $t_{1/2}$ and log a (abscissa), a being the initial concentration of A in the reaction. $A \rightarrow$ Product is depicted in figure.



The rate law is $-\frac{d[A]}{dt} = k[A]^n$

The value of n will be

5. For the reaction $B^+ + H_2O \rightleftharpoons BOH + H^+$, kb for BOH is 1×10^{-6}. If the value of hydrogen constant is 1×10^{-x}. Then value of x is

Rough Work

Mathematics PART – III

SECTION – A

Straight Objective Type

This section contains 10 multiple choice questions numbered 1 to 10. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. If x = 2 is a real root of cubic equation $f(x) = 3x^3 + bx^2 + bx + 3 = 0$ then the value of $\lim_{x \to \frac{1}{2}} \frac{e^{f(x)} - 1}{2x - 1} =$

(A) $\frac{27}{4}$ (B) $\frac{-27}{4}$

(C) $\frac{-27}{8}$ (D) $\frac{27}{8}$

2. If $x \in [-20°, -5°]$ such that $f(x) = \tan(50° + x) + \cos(50° + x) + \cot(50° + x)$ then global maximum value of $f(x)$ is

(A) $\sqrt{\frac{121}{12}}$ (B) $\sqrt{\frac{123}{13}}$

(C) $\sqrt{\frac{121}{4}}$ (D) $\sqrt{\frac{119}{12}}$

3. The number of solution of equation $|z| + |z - 1| + |z - 2| + |z - 3| = 4$ and $z\bar{z} - z - \bar{z} = \frac{5}{4}$ where z is complex number is

(A) zero (B) one

(C) two (D) infinite

Rough work

4. Let us consider a real values bijective function g(x) such that
$g'(x) = \sin^2(x + \pi/4) + 2\cos(x - \pi/4)$ and $g(\pi/4) = 3$ then value of $\left(g^{-1}\right)'$ (3) is
(A) 1 (B) 2
(C) 3 (D) none of these

5. $\int_0^{102}(x-1)(x-2).....(x-100)\left(\frac{1}{x-1}+\frac{1}{x-2}+.....\frac{1}{x-100}\right)dx = \underline{|}p - \underline{|}q$ then p + q is equal to
(A) 203 (B) 201
(C) 202 (D) 204

6. Let ellipse $4x^2 + 16y^2 = 64$ and circle $x^2 + y^2 = r^2$ have common tangent touching at A and B respectively. Maximum length of AB can be
(A) 4 (B) 3
(C) 2 (D) 5

7. f(x) is defined for $x > -1$ and has a continuous derivate. If f satisfies $f(0) = 1$, $f'(0) = 0$;
$(1 + f(x))\, f''(x) = 1 + x$. If x is positive then $f'(x)$ is
(A) always positive (B) always negative
(C) always non-negative (D) none of these

8. Let $f(x) = x^4 - x^3 - 16x^2 + 4x + 48$ then the value of P so that equation $f(|x| + p) = 0$, have 8 real and distinct zeros
(A) $P > -3$ (B) $P > 3$
(C) $P < -3$ (D) $-3 < P < 3$

9. a, b, c denote respectively the sides BC, CA, AB of a triangle ABC. Given that the quadratic equations $ax^2 + bx + c = 0$ and $15x^2 + 8x + 17 = 0$ have the same set of roots, the value of $\frac{r_1 + r_2 + r_3 - r}{R}$ where R denotes the circumradius, r denotes the radius of the incircle and r_1, r_2, r_3 denote the radii of the ex–circles of triangle ABC is
(A) 2 (B) 3
(C) 5 (D) 4

Rough work

10. T_1 and T_2 are the tangents to the hyperbola $x^2 - 2y^2 = 16$ that are perpendicular to the line $x - y + 10 = 0$. Find the perpendicular distance between T_1 and T_2
(A) 1 (B) 4
(C) 6 (D) none of these

Multiple Correct Answer(s) Type

This section contains **5 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE are correct**.

11. Consider the graph of the function $y = e^{\ln\left(\frac{x+3}{x+1}\right)}$ then which of the following is correct
(A) Range of the function is $(1,\infty)$ (B) $f(x)$ has no zeroes
(C) Graph lies completely above the x-axis (D) Domain of f is $(-\infty,-3)\cup(-1,\infty)$

12. $P(x)$ is a fourth degree polynomial such that
(1) $P(-x) = P(x) \forall x \in R$,
(2) $P(x) > 0 \ \forall x \in R$,
(3) $P(0) = 1$,
(4) P(x) has exactly two local minima at x_1 and x_2 such that $| x_1 - x_2 | = 2$.
The line y = 1 touches the curve at a certain point Q and the enclosed area between the line and the curve is $\frac{8\sqrt{2}}{15}$. Let g(x) = Ax^2 + Bx + C ($A \neq 0$) such that $\lim_{x\to 0}\frac{P(x)-g(x)-g(-x)}{x^2}$ is finite and is equal to the slope of the tangent of g(x) at x = -1. Also, P(x) and g(x) have common tangent at Q parallel to x-axis.
(A) the value of A is $-\frac{1}{2}$ (B) the value of B + C is $-\frac{1}{2}$
(C) the value of A + C is 1 (D) the value of A + B + C is –1

Rough work

13. Let S represents a circle of minimum radius in Argand plane which passes through z_1 and z_2, then

(A) Equation of S is $(z-z_1)(\bar{z}-\bar{z}_2)+(z-z_2)(\bar{z}-\bar{z}_1)=0$

(B) Centre of S is represented by (z_1+2z_2)

(C) Radius of S is $\frac{|z_1-z_2|}{2}$

(D) Let z be any complex number lying on circle S then $\frac{(z_2-z)}{|z_2-z|}=\pm\frac{(z-z_1)}{|z-z_1|}i$

14. Let $S_1 = x^2-10x+y^2+16=0,$ and S_2 be the image of S_1 w.r.t. the line x – y = 0 and S_3 be the image of S_2 w.r.t. y = 0. Let S′ is the circle which cut all three circles orthogonally and S′′ is the circle of minimum radius which contain all three circles then

(A) equation of circle S′ is $x^2 + y^2 = 16$ (B) equation of circle S′′ is $x^2 + y^2 = 64$

(C) the radius of S′ is 4 (D) the radius of S′′ is 8

15. A set of lines $x + y – 2 + \lambda_1 (2x + y – 3) = 0$ represents incident rays on an ellipse S = 0 and $2x + 3y – 23 + \lambda_2 (2x – y – 3) = 0$ represents the set of reflective rays from the ellipse where $\lambda_1, \lambda_2 \in R$. If P(3, 7) is a point on the ellipse normal at which meets the major axis at N.

(A) Eccentricity of ellipse is $\frac{\sqrt{5}}{2\sqrt{2}+1}$

(B) N divides line segment joining two foci in the ratio $2\sqrt{2}$: 1

(C) Area of triangle formed by point P and two foci 5

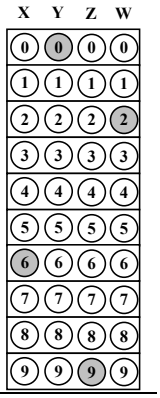
(D) Eccentricity of ellipse is $\frac{\sqrt{5}}{2\sqrt{2}-1}$

Rough work

SECTION – C

Integer Answer Type

This section contains 5 questions. The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown.



1. If circum radius of ΔABC is 3 cm and its area is 6 cm^2 and DEF is triangle formed by foot of perpendicular drawn from A, B, C on sides BC, CA, AB respectively then perimeter of Δ DEF in cm is _____.

2. Let A and B are two sets such that
   A = {x : f(x) = $[x^2]$, ([] denotes GIF) is discontinuous in [0, 2] }
   B = {x : f(x) = $[\tan^2 x]$ $[\cot^2 x]$ ([.] denotes GIF) is non differentiable in (0, π)}
   If m be the total number of onto function from A to B then total number of divisors of m is _____.

3. If two lines x + y = |a| and ax = y + 1 intersect at a point which lies in fourth quadrant then minimum length of focal chord of the parabola $y^2 = 4a^2x + 4|a-1|x + 5$ is _____.

4. $\sum_{r=0}^{9} {}^{20}c_{2r} \; {}^{20}c_{2r+2} = \frac{a}{a+1}\left({}^{39}c_{21} - {}^{19}c_{10} \right)$
   and xy = 4a then total number of ordered pair solution (x, y) where x, y $\in I^+$ is _____.

5. Let n be the number of triangles formed by the line segment of lengths 2, 3, 4, 5, 6, 7. If x and y are two positive integers whose LCM is (n + 2) then total no of ordered pair solution (x, y) will be _____.

Rough work